

# Assignment 2e: Design Check-in (3x4)

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## Section A

## Tasks Review

### Old Tasks

#### Easy - Medication reminder and summary

Taylor is a software engineer in his mid 30s and has had chronic shoulder pain for the last 4 years from a car accident. He has tried multiple approaches to manage his condition, such as pain medication and physical therapy. Those treatments work most of the time. However, recently he has been taking pain medication on and off because of his work schedule shift. The type of medication prescribed to him requires a specific timing, and it's usually during his work time, so he frequently forgets to take the meds. Such inconsistency causes flare-ups for him. He is looking for an integrative application that could remind him to take meds at specific times and make it easier to review which period he has been consistently taking them. Therefore, he could present those results to his provider to adjust treatment plans.

#### Medium - Log symptoms, normally and during flare-ups

Elizabeth is a 21-year-old, recently diagnosed with POTS. She wants to learn how her condition affects her day-to-day by collecting and contextualizing long-term data. She decides to track which days she works out and walks, the daily high temperature, and how she feels each day on a scale of 1-10. She enjoys logging her data, but her main goal is to identify trends and patterns. She likes that she can define what a 1 through 10 day looks like for her, since previous scales, such as the 1-10 pain scale never really made sense to her. She doesn't like filling out long forms, so decides to only track these few things, instead of also tracking medication, stress, sleep, etc. On bad days, she often does not feel like logging, so she goes back and logs her information retroactively when she feels like returning.

#### Hard - Communicating full impact of conditions to their doctors

Rachel has spent years dealing with chronic nerve pain for the past three years. Throughout that time, she has seen multiple doctors from multiple fields of medicine and have gotten different diagnoses. However, none of these diagnoses have actually fully explained her symptoms. She leaves appointments frustrated and misunderstood, wondering if these diagnoses are due to her not being able to communicate her experience clearly enough. She has an upcoming appointment with a new specialist and hopes to finally get an accurate diagnosis. She tries to prepare for it by recalling her symptoms, when they started, and what seems to trigger them. But, she has a hard time describing her experience using standard pain scales because her pain varies widely in intensity and how it affects her daily ability to function. She is worried that if she misses important details, she may be misdiagnosed again.

## New Tasks

### Easy - View Trend/Trigger Analyses, Predictions, and Recommendations

Maya is a 20-year-old college student managing chronic abdominal pain and fatigue. Lately, she has been feeling discouraged because her crashes seem to happen at random, forcing her to cancel plans with friends and fall behind on her coursework. She suspects her diet and sleep patterns are contributing factors, but she can't pinpoint the exact link amid the stress of school. On a Tuesday morning, Maya wants to know whether she is at high risk of a flare-up later in the week, so she can decide whether to front-load her homework or ask for an extension now. She looks for an analysis of her recent data to see if her late-night study sessions or recent meals have historically preceded a crash. By understanding these correlations and seeing a forecast of and recommendations for her energy levels, Maya can make an informed choice to skip a social outing or adjust her diet to avoid a total physical collapse by Friday.

### Medium - Explain conditions to friends, professors, or employees

Kristin deals with unpredictable pain and low energy that makes it difficult for her to follow through on her commitments. After missing a deadline due to a flare-up, Kristin needs to explain her situation to her professor. She struggles to describe how her condition affects her ability to complete work on time, especially because her symptoms are inconsistent and invisible. She wants to communicate her needs clearly so that she can be understood and receive appropriate flexibility. She also wants to be able to communicate her condition to her school friends, which helps them understand why her availability changes and what kind of support she needs, without having to repeatedly justify herself or feel like she's letting others down.

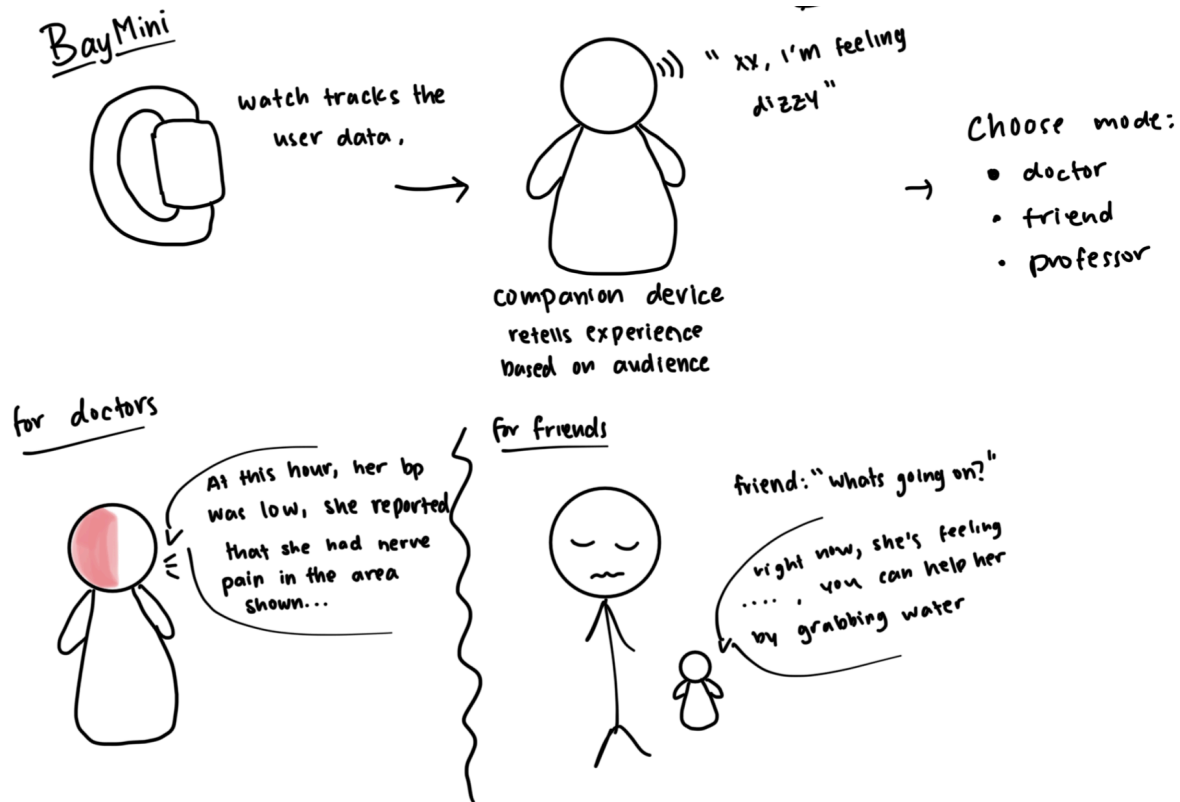
### Hard - Find personalized doctor (health summary), medical, and community resources

Chloe has spent years dealing with doctors who do not understand her specific symptoms. She needs to find a specialist who is known for treating her exact condition rather than just seeing whoever is available. To prepare for a visit, she wants to create a summary of her health data that shows how her pain actually stops her from doing things like painting or working in her lab. She also feels lonely and wants to find a community of people who have the same illness so she can share advice and feel less isolated. Finding the right doctor and a supportive group might finally help her get the right care.

# Designs

## Design 1 - Non-Mobile App - Claudia + Yi

Wearables such as smart watches or smart rings that can track the user's metrics and things like heart rate. Then, the wearable is paired with a companion device Baymini (named after Baymax from the movie Big Hero 6) that the user can verbally talk about their experience with. So in cases of flare ups, the user can log their symptoms by telling the device what exactly they're feeling. The device can then use exact time stamps to compile symptoms and match it with the metrics from the wearable. Then, based on who the audience is, the device can retell the user's experience in a way that would be relevant for the audience.



### Task 1: Log symptoms, normally and during flare-ups

During the presence of symptoms and flare-ups, users could log their symptoms through conversing with the companion device about their conditions. For instance, they could say, "Hey BayMini, I'm feeling xxxx". Then the device will simply log how you're feeling, extract the biometric data at that exact time stamp through the wearable sensor, and compile it together.

### Task 2: Communicating full impact of conditions to doctors

There are different modes, such as doctor, friend and professor, provided by a companion device depending on the context. Different modes have different levels of explanations, meaning

that the language used and tone will be adjusted accordingly. In this case, if we want to communicate and explain the conditions to a doctor, the doctor mode will be chosen. The companion device can deliver the health summary in a curated and professional manner, including proper metrics and medical terms.

### Task 3: Medication reminder and summary

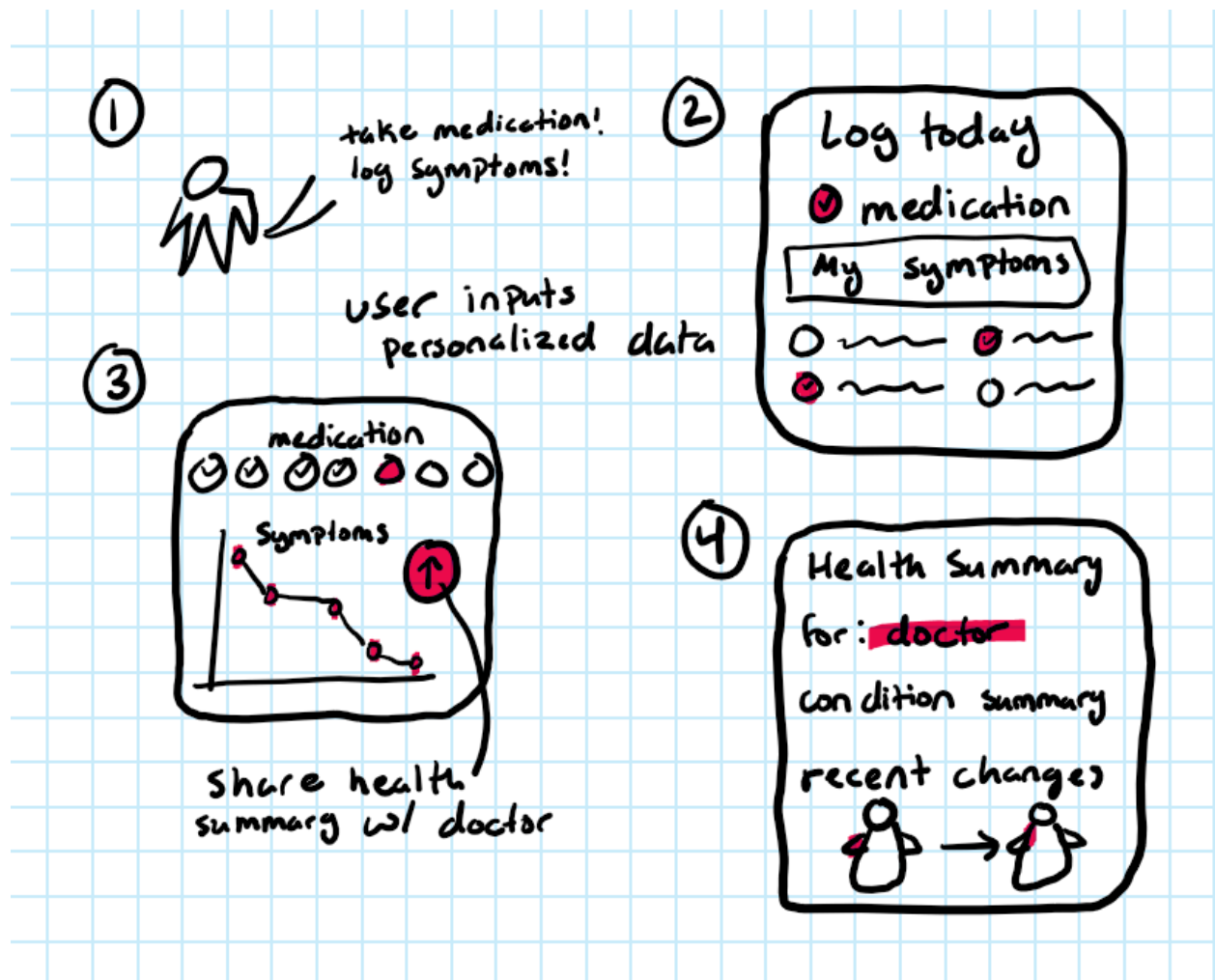
The companion device can deliver verbal and visual reminders to users at when they need to take their medications. For instance, it might say, “Hi, it’s your time to take xxx[meds]”. Apart from that, the device also gives a verbal summary of information that pertains to their condition at the end of a day by activation, such as what meds they take that day, the pattern of their heart rate anomaly and such.

### Task 4: Explain conditions to friends, professors, or employees

When other audience modes are chosen, the device can retell a story in a way that makes sense for the audience. For example, for a friend, it can share about how the user is feeling dizzy and needs water or potentially why they can’t meet up. In addition, when the user is suddenly experiencing a flare-up, it can tell the people around them what is happening and what the user needs or what people around the user can help with.

## Design 2 - Computer/Mobile App - Anna + Kevin

This web app is accessible via mobile and desktop and functions as an investigative hub for lifestyle management. It focuses on correlating daily habits with health outcomes, allowing users to identify trends and triggers for their chronic condition. By integrating diverse data points, from diet and activity to environmental factors and even the user's calendar, the app helps users uncover hidden links between their lifestyle choices and their chronic symptoms.



### Task 1: Medication reminder and summary

The web app will have a weekly medication tracker widget on the main screen that sends notifications and email alerts. Users can mark doses as taken in one click and view weekly summaries of their medication timing/frequency. The system also logs the exact time to help identify if medication timing affects symptom relief later.

## Task 2: Log symptoms, normally and during flare-ups

Users can record their symptoms using both structured inputs, like intensity scales and symptom tags, and unstructured data, such as voice notes or free-form text descriptions. This allows for quick data capture during flare-ups while also providing the space to record more details when the user has more energy.

## Task 3: View Trend/Trigger Analyses, Predictions, and Recommendations

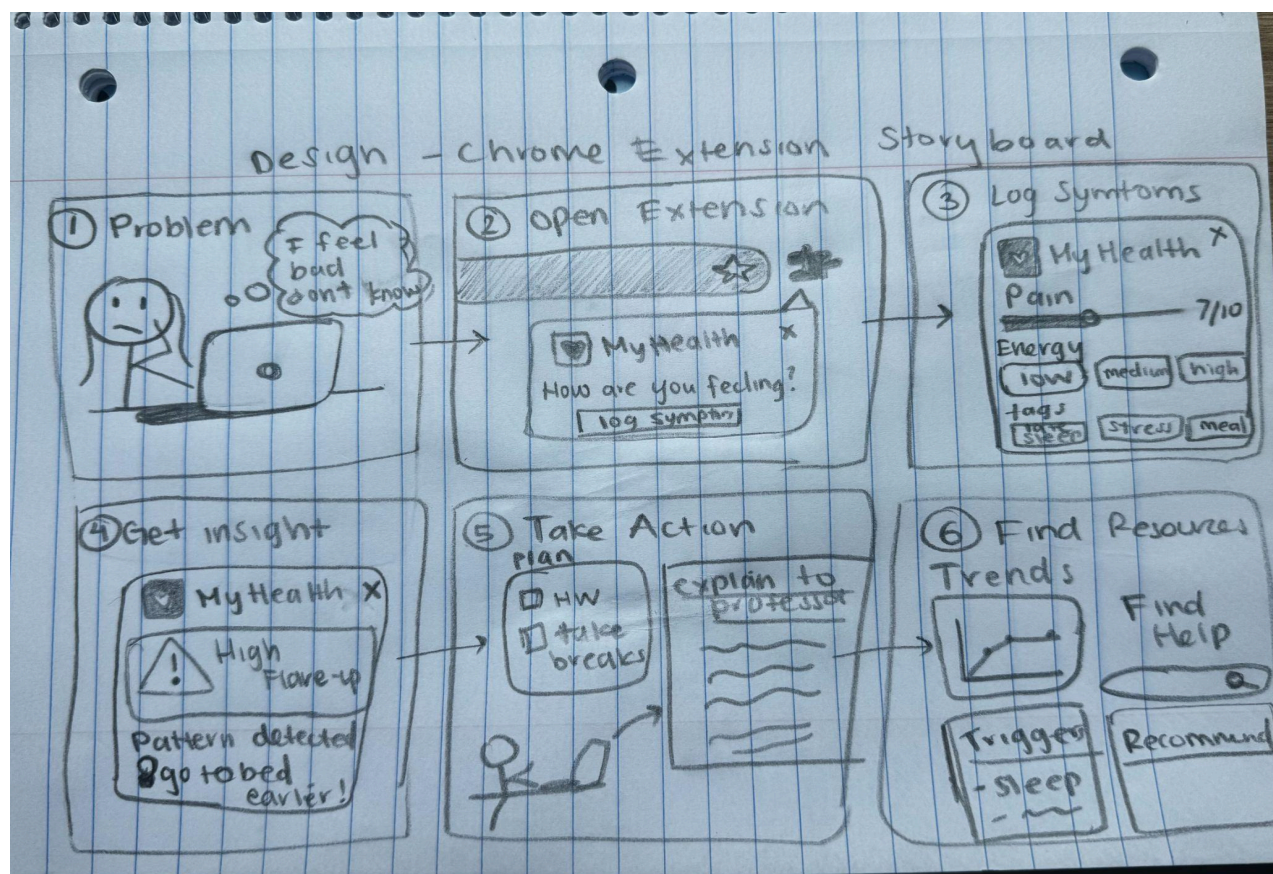
The app will have a tab with visualizations such as heatmaps and correlation charts to show how lifestyle factors like sleep, diet, or stress impact symptom frequency. Based on these trends, the app provides personalized recommendations such as suggesting a change in activity level when certain triggers are detected.

## Task 4: Communicating full impact of conditions to their doctors

Users can generate a PDF that translates their lifestyle data and symptom trends into medical-grade summaries. This provides doctors with more insight on how specific lifestyle variables correlate with the patient's condition.

## Design 3 - Chrome Extensions - Sahana

This Chrome extension is designed to support individuals with chronic conditions by integrating directly into their daily browsing experience. Instead of requiring users to open a separate app, the extension allows them to quickly log symptoms, receive real-time insights, generate an explanation for others, and access relevant resources all within their browser. The goal is to reduce effort, improve tracking consistency, and provide support in moments when users are already working or experiencing symptoms.



### Task 1: View Trend/Trigger Analyses, Predictions, and Recommendations

When a user logs their symptoms through the extension popup, the system analyzes recent patterns in their data, such as sleep habits, stress levels, or activity. Based on this, the extension provides immediate feedback, such as identifying potential triggers or predicting a high risk of a flare-up. These insights are displayed directly in the pop-up, allowing users to quickly understand their condition and make proactive decisions without leaving their current tasks.

## Task 2: Log symptoms, normally and during flare-ups

The extension enables quick, low-effort symptom logging via a small pop-up interface accessible from the browser toolbar. Users can input key information such as pain level, energy, and optional tags in just a few seconds. Because logging is fast and embedded into their workflow, users are more likely to consistently record symptoms-even during flare-ups or busy moments. The design also allows users to log retroactively when needed.

## Task 3: Explain conditions to friends, professors, or employees

The extension provides a feature that helps users generate clear and concise explanations of their condition. When a user is writing an email or message, they can use the extension to auto-generate a message that communicates their situation and needs. This reduces the emotional and cognitive burden of repeatedly explaining an invisible and unpredictable condition, while helping others better understand and respond appropriately.

## Task 4: Find personalized doctor (health summary), medical, and community resources

While users browse the web, the extension provides context-aware recommendations for relevant specialists, medical resources, and support communities. These suggestions are based on the users' logged symptom data and identified patterns. Additionally, the extension compiles long-term data into summaries that users can reference when preparing for doctor visits, helping them communicate their experiences more effectively and find better care.